

华中科技大学

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特征选择策略：SHAP 值与基于重要性方法的比较分析

1.摘要

在处理高维信用卡欺诈数据的背景下,研究人员和从业者通常利用特征选择技术来提升欺诈检测模型的性能。本研究比较了使用 SHAP (Shapley Additive exPlanations) 值和模型内置特征重要性列表所选出的最重要特征对模型性能的影响。两种方法都对特征进行排序,并选择最显著的特征用于模型评估。为了评估这些特征选择技术的有效性,使用了五个分类器构建分类模型: XGBoost、决策树、CatBoost、极端随机树和随机森林。采用精确率-召回率曲线下面积 (AUPRC) 作为评估指标。所有实验均在 Kaggle 信用卡欺诈检测数据集上执行。实验结果表明,基于重要性值的特征选择方法在分类器和各种特征子集大小上均优于基于 SHAP 值的方法。对于在较大数据集上训练的模型,建议使用模型的内置特征重要性列表作为主要的特征选择方法,而不是 SHAP。这一建议基于以下理由: 计算 SHAP 特征重要性是一项独立的活动,而模型在训练过程中自然提供内置特征重要性,无需额外计算。因此,对于较大的数据集和更复杂的模型,选择模型的内置特征重要性列表可以提供一种更高效和实用的方法。

关键词: 特征选择; 类别不平衡; 信用卡欺诈; SHAP; 特征重要性

2.引言

检测信用卡欺诈在金融行业至关重要,并且严重依赖于交易数据集中存储的信息。然而,由于数据质量直接影响建模和分析过程中的决策,金融领域和机器学习面临着一个重大的研究挑战。为了解决这个问题,我们深入研究可用的特征空间,提取一组相关的特征。这突显了在任何建模过程之前,特征选择作为关键数据清洗步骤的重要性。特征选择已广泛应用于数据挖掘和机器学习的各个领域,其目标是移除分析中不相关或冗余的特征。这不仅能够加速模型训练,还能提升分类器性能。

本研究深入比较了两种特征选择方法: 基于 SHAP (Shapley Additive exPlanations) 值的选择和常用的基于重要性的选择。SHAP 利用博弈论概念分两步计算特征重要性: 首先使用所有特征训练一个分类模型,然后计算每个特征的 SHAP 值,接着对它们进行排序以识别对目标问题建模最重要的特征。另一方面,基于重要性的选择在模型训练过程中计算所有特征的重要性。这两种方法都属于嵌入式方法,因为它们都涉及模型构建过程。在我们的特征选择过程中,我们使用了五种学习器: 极限梯度提升、决策树、CatBoost、极端随机树和随机森林。选择这五种学习器是基于它们在模型构建过程中能够生成重要性排序列表的能力。LightGBM 由于其初步结果相较于其他学习器表现不佳,未被纳入我们的选择。我们将基于 SHAP 值的方法命名为 SHAP-XGBoost、SHAP-DT、SHAP-CatBoost、SHAP-ET 和 SHAP-RF,

而将基于重要性的方法简称为 XGBoost、DT、CatBoost、ET 和 RF。总共有 10 种特征选择方法，每类各五种。

为了进行研究，我们专注于 Kaggle 上提供的匿名金融交易数据集——信用卡欺诈检测数据集。该数据集是唯一公开可用的大型信用卡欺诈分析数据。因此，本研究的范围仅限于一个数据集。该数据集包含 284,807 笔交易和 30 个独立特征，其中仅有 492 笔（0.172%）记录被标记为欺诈。使用两种不同的特征选择方法，我们评估了五组分类器模型的性能，这些模型使用了不同特征选择技术（SHAP-XGBoost vs. XGBoost, SHAP-DT vs. DT, SHAP-CatBoost vs. CatBoost, SHAP-ET vs. ET, SHAP-RF vs. RF）及其各自选择的特征。根据各自的分数，我们选择了排名前 3、5、7、10 和 15 的特征。对于分类，我们使用五个分类器（与特征选择中使用的模型相同）构建信用卡欺诈检测模型。使用精确率-召回率曲线下面积（AUPRC）来评估分类器，并且我们还进行了显著性水平为 $\alpha=0.01$ 的统计检验，以评估结果的统计显著性。

据我们所知，本研究是首次在欺诈检测乃至机器学习其他潜在应用领域，对基于 SHAP 值的特征选择和基于重要性的特征选择的性能进行全面实证比较的研究。

本文的其余部分组织如下。我们首先概述相关工作，以展示我们在此展示的研究工作的新颖性。接着我们介绍实验中使用的的方法，包括对两种特征方法、分类器、交叉验证和性能指标的说明。然后我们描述数据集、实验设计和实验结果。最后，我们总结了本研究的关键亮点，并对未来工作提出建议。

3.相关工作

特征选择是各种数据挖掘和机器学习应用中广泛使用的技术。其主要目标是识别能够最小化分类器预测误差的特征子集。在本研究中，我们全面回顾了使用 SHAP (Shapley Additive exPlanations) 值或模型内置特征重要性列表进行特征选择的研究文献。我们发现，在信用卡欺诈检测数据集的背景下，仅有有限数量的研究使用了模型内置特征重要性列表进行特征选择，但我们没有发现任何研究专门使用 SHAP 进行信用卡欺诈检测的特征选择。相反，我们发现了一些在其他应用领域使用 SHAP 进行特征选择的研究。此外，我们没有发现任何研究直接比较了使用 SHAP 特征重要性所选特征构建的模型与使用内置特征重要性所选特征构建的模型的性能。因此，我们的研究在信用卡欺诈检测领域做出了独特贡献，因为它探索了 SHAP 与模型内置特征重要性列表在特征选择方面的比较，这一视角在现有文献中尚未得到广泛探索。

Rtayli 和 Enneya 应用了一种有监督的特征选择方法——随机森林，来识别最具预测性的特征。随机森林（RF）是一种集成学习算法，通过装袋法并行训练。近年来，随机森林越来越多地被用作特征选择方法，因为它能够处理复杂、高维的数据集，并能检测特征间的交互。它还降低了过拟合的风险（当模型过于复杂并过于紧密地拟合训练数据时会发生）。此外，随机森林通过测量使用某个特征进行节点分割时节点不纯度的减少来计算特征重要性。不纯度减少得越多，该特征被认为越重要。通过根据重要性对特征排序，随机森林可以帮助选择与分类任务最相关的特征。从信用卡欺诈检测数据集中选择特征子集后，作者运行了支持向量机来查找欺诈交易。该模型达到了 95.12% 的准确率、87% 的敏感性和 0.91 的 AUC，优于其他三个模型（隔离森林、决策树和局部异常因子）。该研究未提供关于所选特征数量的明确信息。此外，作者没有对所选特征与使用所有可用特征之间的性能进行比较。再者，值得注意的是，对于不平衡数据的分类，使用 AUC 作为指标已被发现具有误导性。

Rosley 等人 在他们使用信用卡欺诈检测数据集的研究中，首先过滤掉 z 分数大于或等于 3 的数据，然后使用最小-最大缩放对剩余数据进行归一化。接着他们使用 Boruta 计算每个特征的重要性分数。Boruta 是一种有监督的特征选择算法，被设计为围绕随机森林分类器的包装器，用于识别数据集中的重要特征。他们保留了重要性分数为 0.5 或更高的特征来训练每次迭代的自编码器。该模型通过定义重建误差的阈值来标记交易为合法或欺诈，从而检测信用卡欺诈。然而，预处理步骤中选择的特征数量未由作者具体说明。作者使用准确率、精确率、召回率和 F1 分数评估模型。在处理类别严重不平衡的数据集时，由于多数类占压倒性多数，这些指标可能并不合适。

Waspada 等人使用随机森林分类器计算每个特征的重要性分数。重要性分数低的特征被丢弃。论文列出了所有特征的重要性分数。作者分析了影响隔离森林模型检测信用卡交易欺诈性能的几个因素（数据集分割比例、前 k 个特征的选择、训练数据中欺诈数据的数量以及超参数值的设置）。隔离森林是一种流行的无监督异常检测方法。他们的研究表明，通过设置训练-测试比例为 60:40，使用前五个特征（V14、V4、V17、V12、V11），仅使用 60% 的欺诈数据，并设置超参数为树数量 100、最大样本数 128、污染率为 0.001，可以获得最佳结果。该模型显示出令人印象深刻的结果：精确率为 80.7143%，召回率为 76.3514%，F1 分数为 78.4722%，接收者操作特征曲线下面积为 0.97371，精确率-召回率曲线下面积为 0.759228。Waspada 等人仅使用了单一基于重要性的特征选择方法，未将 SHAP 纳入特征选择，而我们在研究中已实现。

Liu 等人在他们的研究中，利用 SHAP 对 UCI 帕金森病医疗数据集进行特征选择。他们将 SHAP 值与四种分类器结合使用：深度森林（gcForest）、极限梯度提升（XGBoost）、轻量级梯度提升机（LightGBM）和随机森林（RF）。每个分类器都用于计算各个特征的 SHAP 值。为了评估 SHAP 特征选择的有效性，他们将其与三种基于过滤器的特征选择方法（Fscore、方差分析和互信息）进行了比较。实验采用 70:30 的训练和测试比例进行，并且特征选择应用于训练数据集。结果表明，基于 SHAP 值特征选择的 gcForest 模型取得了令人印象深刻的 91.78% 的分类准确率和 0.945 的 F1 分数，选择了 150 个特征。这一性能超过了他们研究中考虑的其他特征选择方法的结果。虽然作者在训练数据集上专门使用了基于 SHAP 值的特征选择，但我们则在整个数据集上使用了 SHAP 方法，随后在特征选择程序后进行了交叉验证。

Marcilio 和 Eler 采用 SHAP 方法作为特征选择技术，并与三种广泛使用的特征选择方法（互信息、递归特征消除和 ANOVA）进行了比较。SHAP 过程涉及使用 XGBoost 作为基础模型。他们使用 XGBoost 分类器在五个 UCI 数据集上和使用 XGBoost 回归器在另外三个 UCI 数据集上进行了实验。他们的研究表明，就接收者操作特征曲线下面积指标而言，SHAP 优于三种常用方法。然而，观察到 SHAP 需要比其他特征选择方法更多的计算时间。值得注意的是，Marcilio 和 Eler 实验中使用的数据集不是高度不平衡的，也不在信用卡欺诈领域。此外，与引起我们注意的 Kaggle 信用卡欺诈检测数据集相比，这些数据集的规模明显更小。

在我们对文献的回顾中，我们发现仅采用了单一的特征选择方法，无论是基于 SHAP 值还是基于重要性。值得注意的是，尚未发现任何研究比较这两种方法，尤其是在信用卡欺诈检测领域。为了填补这一空白，我们的研究使用每种方法中的五种学习器，对这两种特征选择方法进行了比较分析。

4.方法

4.1. 基于重要性的特征选择方法

基于重要性的特征选择方法利用决策树从给定的数据集中识别相关特征。这些基于决策树的分类器，如极限梯度提升（XGBoost）、极端随机树（ET）、随机森林（RF）、CatBoost 和决策树（DT），在监督机器学习模型拟合过程中，具备确定特征重要性的内置能力。因此，它们可以根据特征在分类任务中的重要性对特征进行排序，使其在特征选择中很有价值。通过丢弃相关性较低的特征并保留最重要的特征，可以创建更高效和准确的模型。

在本研究中，采用了五种基于重要性的特征选择方法：XGBoost、决策树（DT）、CatBoost、

极端随机树（ET）和随机森林（RF）。

XGBoost 和 CatBoost 作为广泛使用的梯度提升算法脱颖而出，各自采用不同的方法来计算特征重要性分数。虽然两种算法都构建决策树集成，但它们推导特征重要性分数的方法有所不同。在 XGBoost 中，这些分数使用“增益”方法计算，评估每个特征在整个提升过程中对模型性能的影响。相比之下，CatBoost 的决策树集成基于特征被用于分割的频率以及通过这些分割实现的模型性能提升来计算特征重要性。

决策树分类器是一种用于分类任务的机器学习算法。它通过在每个节点根据最具信息性的特征递归地分割数据，构建一个决策及其可能结果的树状模型。决策树通过评估在构建树的过程中特征降低 Gini 不纯度（或增加纯度）的能力来生成特征重要性分数。

极端随机树和随机森林都基于决策树集成，共享 Gini 不纯度和平均不纯度减少等通用原则来衡量特征重要性。然而，极端随机树在树构建的决策过程中引入了更高的随机性。这种额外的随机性可能导致不同的重要性分数，可能影响模型偏差和方差的平衡。

4.2 基于 SHAP 值的特征选择方法

Shapley Additive exPlanation (SHAP) 由 Lundberg 和 Lee 提出，作为一种解释机器学习模型预测的方法而广受欢迎。通过利用博弈论技术，SHAP 提供了对每个特征对特定预测贡献的洞察。它属于一系列加法特征归因技术，这些技术保持模型无关性，使其普遍适用于各种机器学习和深度学习模型。这些技术将重要性归因于单个输入特征，有助于更好地理解模型行为。

在特征选择的背景下，基于 SHAP 的方法的工作流程如下：首先在本研究中的整个数据集上训练分类模型（如 XGBoost 和决策树）。随后，为每个实例计算 SHAP 值，然后将这些值在整个数据集上进行聚合，以得出每个特征的平均绝对值。由于这个过程，SHAP 值的计算变得计算复杂。平均 SHAP 值表示每个特征在整个数据集上对模型预测的典型影响，而绝对 SHAP 值代表特征的重要性，无论其方向（正或负）。通过按照平均绝对 SHAP 值降序对特征进行排序，具有较高 SHAP 值的特征被识别为对模型预测影响更大。

4.3 分类器

在本研究中，使用五个不同的分类器构建信用卡欺诈检测模型，即 XGBoost、决策树（DT）、CatBoost、极端随机树（ET）和随机森林（RF）。在这五种学习器中，XGBoost、CatBoost、ET 和 RF 都是基于决策树的集成分类器。我们选择这些学习器是因为它们能高效处理复杂、高维数据，并且在广泛的分类任务中以其卓越性能而闻名。

XGBoost 和 CatBoost 都是广泛用于机器学习任务的梯度提升框架，尤其是分类。这两种算法被认为是非常有效的并能产生准确的预测。然而，性能可能会因具体数据集和问题而异。XGBoost 是梯度提升决策树集成方法的高级改进版。GBDT 最初由 Friedman 在 2001 年提出。XGBoost 在多个方面增强了 GBDT。首先，它在训练期间使用了一个改进的损失函数，该函数包含一个额外的正则化项，有效防止过拟合。其次，XGBoost 引入了一种“近似算法”来计算构成决策树中的分割，该算法非常适合分布式环境以及整个数据集无法装入主内存的情况。此外，XGBoost 包含一种专门处理稀疏数据的算法，其中大多数值几乎恒定，偶有异常。“稀疏感知分割查找”功能使 XGBoost 能够有效利用稀疏数据。另一方面，CatBoost 以其在处理分类特征和缺失值方面的鲁棒性而闻名，适用于具有此类特征的数据集。CatBoost 的核心算法是有序提升，涉及对决策树使用的实例进行排序。相比之下，XGBoost 依赖于加权分位数草图和处理稀疏性的函数。加权分位数草图是一种近似树学习技术，用于合并和剪枝操作，而稀疏性处理的是零值或缺失值。

Breiman 在 1996 年的一篇论文中引入了机器学习领域中的装袋法概念。由于我们的研究围绕二分类，我们关注 Breiman 关于装袋法应用于二分类的思想。极端随机树和随机森林都是集成学习算法，属于基于决策树的装袋法家族。随机森林由 Breiman 提出。随机森林在装袋法原理的基础上进行了改进。在随机森林中，每棵树都是使用随机选取的特征子集和样本子集构建的。这种随机性有助于降低树之间的相关性并减少过拟合。极端随机树扩展了随机森林的概念，通过随机选择决策树分割的值，这使它们在某些场景中可能更具鲁棒性和计算效率。两者之间的选择通常取决于数据的具体特征以及偏差与方差之间的权衡。我们在此省略这些学习器的详细信息。

决策树是一种广泛使用的监督机器学习算法，主要应用于分类和回归任务。它是一种非线性模型，根据特征值递归地将输入数据分割成子集。决策树中的每个节点代表基于特定特征和阈值的决策，便于根据输入数据的特征值进行预测。生成的决策树结构具有高度可解释性，每个内部节点代表基于特征的决策，边表示结果，叶节点提供预测。

为了确保我们结果的可重复性，我们修改了表 1 中列出的特定超参数设置。此外，我们为所有分类器设置了随机数生成器种子，以确保结果一致且可重复。所有其他设置均保留其默认值。树深度的确定参考了先前文献中记录的实验，旨在实现捕捉数据中复杂模式与缓解过拟合之间的适当平衡。

分类器	参数名称	参数设置
-----	------	------

CatBoost	task_type	GPU*
	max_ctr_complexity	1
	max_depth	5
ET	max_depth	8
XGBoost	max_depth	3
	tree_method	gpu_hist*
随机森林	max_depth	4

*该设置选择分类器的图形处理器实现

4.4 性能指标

为了评估特征选择技术的有效性，我们在特征选择过程之后构建了分类模型。本研究中这些模型的评估基于精确率-召回率曲线下面积指标。

在一个两类分类问题中，例如区分欺诈（正例）和正常（负例）实例，我们会遇到四种可能的预测结果：真阳性（正确分类的正例实例）、假阳性（错误分类为正例的负例实例）、真阴性（正确分类的负例实例）和假阴性（错误分类为负例的正例实例）。

AUPRC 代表精确率-召回率曲线下的面积，该曲线说明了针对特定分类阈值时召回率（真阳性率）和精确率之间的权衡。精确率的定义是：

$$\text{精确率} = \frac{\text{真阳性}}{\text{真阳性} + \text{假阳性}}$$

召回率或真阳性率的定义是：

$$\text{召回率} = \frac{\text{真阳性}}{\text{真阳性} + \text{假阴性}}$$

为了计算 AUPRC，我们绘制许多分类阈值下的精确率与召回率的关系图，然后计算曲线下面积。较高的 AUPRC 值表示模型性能更优。AUPRC 范围从最小 0 到最大 1。

4.5 交叉验证

交叉验证是一种技术，用于在不重复使用相同数据的情况下训练和测试机器学习模型。该过程涉及以相对平衡的方式将数据集划分为预定数量的子集或折。在本研究中，我们使用了五折交叉验证，其中每一折用作测试数据，而其余四折指定为训练数据。为了最小化因偶然或不利的分割而产生的潜在偏差，我们独立运行了十次五折交叉验证。

值得注意的是，出于可重复性的考虑，特征选择过程与交叉验证步骤是分开进行的。换句话说，特征选择程序是在原始数据集上执行的。

5.实验

5.1. 数据集

本研究中进行的实验使用了可从 Kaggle 网站下载的信用卡欺诈检测数据集。该数据集包含匿名的金融交易，具体是 2013 年 9 月两天内欧洲持卡人进行的信用卡交易。如前所述，在总共 284,807 笔交易中，有 492 笔是欺诈交易，导致数据集不平衡，只有 0.172% 的交易是欺诈性的，其余被视为正常或非欺诈交易。

信用卡欺诈检测数据集有 30 个数值输入特征，其中 $V_1, V_2, \dots, V_{28}$ 已使用主成分分析进行数值转换，用于数据分析和特征降维。然而，“时间”和“金额”特征未被转换。“时间”特征表示自第一笔交易以来的秒数，“金额”特征代表信用卡交易的金额。为了避免影响结果的可靠性，“时间”特征被排除在分析之外，因为它是一个模型可以记忆的唯一特征。因此，有 29 个输入特征可用于进一步的实验。在输入分类器进行训练或分类之前，特征被归一化到 $[0, 1]$ 范围内。类别特征用于区分合法交易和欺诈交易。在此上下文中，值 1 代表欺诈交易，值 0 表示正常交易。

5.2. 实验设计

在我们的实验中，我们研究了两种不同的特征选择技术：基于 SHAP 值的特征选择和基于重要性的特征选择方法。为了评估特征选择方法的有效性，我们利用特征选择方法选择的特征子集构建了分类模型。分类模型使用五个分类器构建：XGBoost、决策树、CatBoost、极端随机树和随机森林。

我们的实验在一个分布式计算平台上进行，该平台由配备 16 核英特尔至强 CPU、每 CPU 256 GB RAM 和 Nvidia V100 GPU 的节点组成。所有训练和测试程序均使用 Python 编程语言实现。SHAP 作为 Python 编程语言的开源库公开可用。除了用于特征重要性的 SHAP 值外，该库还提供了几种可视化 SHAP 特征重要性值的工具。实验实现使用了 Python 数据科学栈。

首先，我们分别使用十种特征选择方法（SHAP-XGBoost、XGBoost、SHAP-DT、DT、SHAP-CatBoost、CatBoost、SHAP-ET、ET、SHAP-RF 和 RF）对特征进行排序。在特征排序之后，我们选择排名前 3、5、7、10 和 15 的特征，包括类别属性，来构建最终的训练数据集。随后，我们将分类器应用于这些训练数据集，确保模型构建过程中使用的分类器与特征选择中使用的分类器保持一致。我们使用 AUPRC 来评估分类模型的性能。对于每种特征选择方法和分类器，我们总共有 $5(\text{特征子集大小}) \times 10(\text{运行轮次}) \times 5(\text{折}) = 250$ 个 AUPRC 分数。

5.3. 结果与讨论

如前所述，我们引入了十种特征选择方法，即两种特征选择技术与五种分类器的组合。我们展示了从每种方法获得的重要性列表，其中我们关注排名前 15 的最重要特征。重要性由 SHAP 值（对于 SHAP-XGBoost、SHAP-DT、SHAP-CatBoost、SHAP-ET 和 SHAP-RF）或内置重要性分数（对于 XGBoost、DT、CatBoost、ET 和 RF）决定。在表 2、3、4、5、6 中，我们展示了特征排名，其中排名 1 对应最高的 SHAP 值或重要性分数。值得注意的是，当使用不同的训练模型时，SHAP 值可能会有所不同。值得注意的是，在所有十种特征选择方法中，特征 V14 都脱颖而出，成为前三个特征之一。此外，特征 V4 在所有特征选择方法中始终出现，并保持在排名前 15 以内。

表 2 由 SHAP-XGBoost 和 XGBoost 选择的特征；特征按其重要性值从高到低列出

排名	SHAP-XGBoost	XGBoost
1	V14	V17
2	V4	V14
3	V12	V10
4	Amount	V27
5	V8	V12
6	V11	V26
7	V7	V4
8	V10	V1
9	V5	V8
10	V19	V7
11	V26	V16
12	V27	V9
13	V3	Amount
14	V16	V13
15	V18	V3

表 3 由 SHAP-DT 和 DT 选择的特征；特征按其重要性值从高到低列出

排名	SHAP-DT	DT
1	V14	V17

2	V17	V14
3	V12	V27
4	V4	V12
5	V1	V10
6	V20	V26
7	V19	V24
8	V9	V16
9	V10	V7
10	V7	V20
11	V21	V4
12	V26	V1
13	V27	V23
14	Amount	V19
15	V22	V15

表 4 由 SHAP-CatBoost 和 CatBoost 选择的特征；特征按其重要性值从高到低列出

排名	SHAP-CatBoost	CatBoost
1	V1	V1
2	V14	V4
3	V4	V14
4	V8	Amount
5	V26	V11
6	V6	V26
7	Amount	V13
8	V24	V8
9	V12	V17
10	V13	V3
11	V11	V20
12	V18	V18
13	V10	V24

14	V17	V15
15	V19	V28

表 5 由 SHAP-ET 和 ET 选择的特征；特征按其重要性值从高到低列出

排名	SHAP-ET	ET
1	V14	V14
2	V17	V17
3	V12	V12
4	V4	V10
5	V10	V11
6	V11	V16
7	V16	V18
8	V3	V4
9	V9	V9
10	V18	V3
11	V7	V7
12	V19	V2
13	V1	V21
14	V2	V19
15	V15	V26

表 6 由 SHAP-RF 和 RF 选择的特征；特征按其重要性值从高到低列出

排名	SHAP-RF	RF
1	V14	V17
2	V17	V12
3	V12	V14
4	V10	V10
5	V4	V16
6	V1	V11
7	V11	V9
8	V16	V4

9	V2	V7
10	V7	V18
11	V19	V26
12	V3	V21
13	V5	V1
14	Amount	V8
15	V18	V3

AUPRC 方面的分类性能结果如表 7、8、9、10、11 所示。报告的值代表十轮五折交叉验证结果的平均值。结果是通过使用排名前 3、5、7、10、15 的特征以及类别属性创建新数据集形成的最终训练数据获得的。我们对模型对（相同分类器但不同特征选择方法）进行了统计 z 检验，其中每对模型包含一个使用由 SHAP 或模型内置特征重要性列表选择的 n 个最重要特征构建的模型。n 的值范围从 3 到 15。零假设是两个模型的平均 AUPRC 分数之间没有显著差异。在表 7、8 中，“胜者”列根据显著性水平 $\alpha=0.01$ 的 z 检验结果，表明 SHAP 或内置特征选择方法是否具有更高的平均 AUPRC 值。如果均值差异不显著，我们报告为平局。

表 7 SHAP 和 XGBoost 特征选择方法在 AUPRC 分数上的比较

特征数量	SHAP-XGBoost	XGBoost	p 值	胜者
3	0.7247	0.7727	0.0000	XGBoost
5	0.8165	0.7978	0.0121	平局
7	0.8302	0.8255	0.5005	平局
10	0.8446	0.8350	0.0041	SHAP-XGBoost
15	0.8535	0.8557	0.7097	平局

表 8 SHAP 和 DT 特征选择方法在 AUPRC 分数上的比较

特征数量	SHAP-DT	DT	p 值	胜者
3	0.7421	0.7323	0.4968	平局
5	0.7493	0.7414	0.6293	平局
7	0.7594	0.7666	0.7013	平局
10	0.7380	0.7686	0.2429	平局
15	0.7664	0.7564	0.5058	平局

表 9 SHAP 和 CatBoost 特征选择方法在 AUPRC 分数上的比较

特征数量	SHAP-CatBoost	CatBoost	p 值	胜者
3	0.6106	0.7235	0.0000	CatBoost
5	0.7266	0.7745	0.0000	CatBoost
7	0.7897	0.8279	0.0000	CatBoost
10	0.8333	0.8472	0.0000	CatBoost
15	0.8506	0.8491	0.7502	平局

表 10 SHAP 和 ET 特征选择方法在 AUPRC 分数上的比较

特征数量	SHAP-ET	ET	p 值	胜者
3	0.7796	0.7843	0.6756	平局
5	0.8172	0.8118	0.4243	平局
7	0.8143	0.8137	0.9179	平局
10	0.8175	0.8168	0.9152	平局
15	0.8086	0.8048	0.7238	平局

表 11 SHAP 和 RF 特征选择方法在 AUPRC 分数上的比较

特征数量	SHAP-RF	RF	p 值	胜者
3	0.8097	0.8137	0.5673	平局
5	0.8396	0.8248	0.0133	平局
7	0.8416	0.8382	0.6126	平局
10	0.8447	0.8479	0.6399	平局
15	0.8544	0.8512	0.6693	平局

表 7 显示，对于特征子集大小为 5、7 和 15 的 XGBoost 模型，结果是平局。然而，对于特征子集大小 3，p 值小于显著性水平 0.01，表明 AUPRC 分数存在显著差异。因此，对于特征数量为 3 的情况，XGBoost 优于 SHAP-XGBoost。另一方面，对于特征子集大小 10，SHAP-XGBoost 优于 XGBoost。

表 8 表明，对于测试的所有特征数量（3、5、7、10 和 15），SHAP-DT 和 DT 之间的 AUPRC 分数没有显著差异。因此，基于 AUPRC 分数，我们不能宣布两种特征选择方法之间有胜者。表 10 和表 11 与表 8 类似。结果表明，对于给定的数据集和评估指标，在不同的

特征规模下，SHAP 特征选择方法与传统的基于重要性值的决策树、极端随机树或随机森林方法之间没有一致的优势性能。

表 9 展示了 SHAP-CatBoost 和 CatBoost 特征选择方法在不同特征数量下的 AUPRC 分数比较。总之，对于特征数量 3-10，CatBoost 在 AUPRC 方面持续优于 SHAP-CatBoost，且差异具有统计显著性，p 值为 0.0000。然而，对于特征数量 15，两种方法之间没有统计学上的显著差异，结果为平局。

总的来说，两种特征选择方法在各种情况下的性能相当。然而，在某些特定实例中，如某些 XGBoost 和 CatBoost 模型，出现了差异。值得注意的是，当特征子集大小为 3 时，XGBoost 表现优于 SHAP-XGBoost，而 CatBoost 在特征大小为 3、5、7 和 10 时表现优于 SHAP-CatBoost。此外，当特征子集大小为 10 时，SHAP-XGBoost 表现优于 XGBoost。

对 AUPRC 性能指标进行了方差分析，结果报告在表 12 中。分析中考虑了三个因素：大小、分类器和技术。“大小”因素包括特征子集大小 3、5、7、10 和 15，“分类器”因素包括五个分类器，而“技术”因素包括两种特征选择方法：基于 SHAP 值（用 SHAP 表示）和基于重要性值（用 Importance 表示）。统计检验使用了 $\alpha=1\%$ 的显著性水平。ANOVA 结果表明，就 AUPRC 指标而言，每个主因素的组别之间存在显著差异，因为表中最后一列的所有 $\text{Pr}(> F)$ 或 p 值都小于 0.01 的截断值。

表 12 对 AUPRC 性能中作为因素的"大小"、"分类器"和"技术"进行的方差分析

	自由度	平方和	均方	F 值	$\text{Pr}(> F)$
大小	4	1.90	0.48	237.44	小于 10^{-4}
分类器	4	2.11	0.53	262.91	小于 10^{-4}
技术	1	0.05	0.05	24.77	小于 10^{-4}
残差	2490	4.99	0.00		

由于 ANOVA 检验结果显示所有因素都对 AUPRC 分数有显著影响，我们进行了 Tukey's Honestly Significant Difference (HSD) 检验，以根据其对 AUPRC 分数的影响对“技术”和“分类器”进行排序。性能按字母顺序排名，组'a'具有最高的 AUPRC 分数。同一性能组中的项目表示它们之间没有统计学上的显著差异。HSD 检验结果如表 13、14、15 所示。

根据 HSD 检验，很明显，使用大小为 15 和 10 的特征子集进行特征选择，在 AUPRC 方面优于较小的子集大小。这表明构建特征子集大小为 15 或 10 的模型是有利的。较小的特

征集能缩短模型训练时间并改善结果。在五个分类器中，随机森林表现出最高的 AUPRC，其次是 XGBoost 和 ET，而决策树表现出相对较差的性能。表 15 表明，基于重要性值的特征选择方法在所有特征子集大小和学习器上，均显著优于基于 SHAP 值的特征选择方法。

表 13 对"大小"因素进行 ANOVA 后的 HSD 检验分组

组 a 包含：15
组 ab 包含：10
组 b 包含：7
组 c 包含：5
组 d 包含：3

表 14 对"分类器"因素进行 ANOVA 后的 HSD 检验分组

组 a 包含：RF
组 b 包含：XGBoost, ET
组 c 包含：CatBoost
组 d 包含：DT

表 15 对"技术"因素进行 ANOVA 后的 HSD 检验分组

组 a 包含：Importance
组 b 包含：SHAP

如前所述，SHAP 是一个外部工具，SHAP 特征选择的计算时间取决于几个因素，包括模型的复杂性、特征数量、数据集大小以及需要计算 SHAP 值的实例数量。计算 SHAP 值的复杂性通常高于其他特征重要性方法（如基于决策树的分类器）。因此，我们得出结论：对于具有大量特征和大型数据集的模型，使用内置特征重要性来选择特征子集可能更合适。

6.结论

处理机器学习中的高维性问题对模型性能评估有显著影响。本研究特别集中于比较两种特征选择技术：通过 SHAP 值识别最关键的特征以及依赖模型的内在特征重要性列表。使用信用卡欺诈检测数据集，我们生成了多个训练数据集。我们采用五个分类器和不同的特征子集大小，将两种特征选择方法应用于每个分类器。我们的结果表明，总体而言，在本研究使用的分类器和各种特征子集大小上，基于重要性值的特征选择方法优于基于 SHAP 值的方法。

然而，在 XGBoost 模型中出现了一些显著的变化。对于特征子集大小为 3 的情况，XGBoost 优于 SHAP-XGBoost，而对于特征子集大小为 10 的情况，SHAP-XGBoost 优于 XGBoost。在 CatBoost 的情况下，对于小于 15 的特征数量，CatBoost 优于 SHAP-CatBoost。

值得注意的是，计算 SHAP 特征重要性在实验方法中引入了一个额外的步骤。根据我们的发现，实施 SHAP 的投资回报率可能相对较低，尤其是在内置特征选择方法可用时，对于大型数据集尤其如此。此外，与 SHAP 相关的可观计算开销可能使其在处理大数据时不切实际。对于未来的研究，我们计划探索这两种特征选择方法在不同应用领域中的表现。

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RESEARCH

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Feature selection strategies: a comparative analysis of SHAP-value and importance-based methods

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Abstract

In the context of high-dimensional credit card fraud data, researchers and practitioners commonly utilize feature selection techniques to enhance the performance of fraud detection models. This study presents a comparison in model performance using the most important features selected by SHAP (SHapley Additive exPlanations) values and the model's built-in feature importance list. Both methods rank features and choose the most significant ones for model assessment. To evaluate the effectiveness of these feature selection techniques, classification models are built using five classifiers: XGBoost, Decision Tree, CatBoost, Extremely Randomized Trees, and Random Forest. The Area under the Precision-Recall Curve (AUPRC) serves as the evaluation metric. All experiments are executed on the Kaggle Credit Card Fraud Detection Dataset. The experimental outcomes and statistical tests indicate that feature selection methods based on importance values outperform those based on SHAP values across classifiers and various feature subset sizes. For models trained on larger datasets, it is recommended to use the model's built-in feature importance list as the primary feature selection method over SHAP. This suggestion is based on the rationale that computing SHAP feature importance is a distinct activity, while models naturally provide built-in feature importance as part of the training process, requiring no additional effort. Consequently, opting for the model's built-in feature importance list can offer a more efficient and practical approach for larger datasets and more intricate models.

Keywords: Feature selection, Class imbalance, Credit card fraud, SHAP, Feature importance

Introduction

Detecting credit card fraud is crucial within the finance industry and heavily relies on the information stored in transaction datasets. However, the finance field and machine learning face a significant research challenge due to the quality of data, as it directly influences decisions made during modeling and analysis [1, 2]. To tackle this issue, we delve into the available feature space, extracting a pertinent set of features. This underscores the importance of feature selection as an essential data cleansing step before engaging in any modeling process. Feature selection has found application in various

contexts within data mining and machine learning, with the goal of removing irrelevant or redundant features from the analysis. This not only results in expedited model training but also enhances classifier performance.

This study delves into a comparison between two feature selection methods: Shapley Additive exPlanation (SHAP)-value-based selection [3] and commonly used importance-based selection [4, 5]. SHAP leverages game theory concepts to compute feature importance in two steps: training a classification model using all features in the initial interaction and then computing SHAP values for each feature, subsequently ranking them to identify the most significant features for modeling the target problem. On the other hand, importance-based selection computes feature importance for all features during the model training process. Both methods are embedded since they involve the model-building process. In our feature selection process, we utilize five learners: Extreme Gradient Boosting (XGBoost) [6], Decision Tree (DT) [7], CatBoost [8], Extremely Randomized Trees (ET) [9], and Random Forest (RF) [10]. The selection of these five learners is based on their ability to generate an importance ranking list during the model-building process. LightGBM [11] was not included in our choices due to its poor performance, as indicated by our preliminary results in comparison to other learners. We have designated the SHAP-value-based methods as SHAP-XGBoost, SHAP-DT, SHAP-CatBoost, SHAP-ET, and SHAP-RF, while referring to the importance-based methods simply as XGBoost, DT, CatBoost, ET, and RF. In total, there are 10 feature selection methods, five from each category.

To conduct our study, we focus on the Credit Card Fraud Detection Dataset, a set of anonymized financial transactions available on Kaggle [12]. This dataset is the only publicly available large data for credit card fraud analysis. Hence the scope of the study is limited to one dataset. With 284,807 transactions and 30 independent features, only 492 (0.172%) records are labeled fraudulent. Using two different feature selection methods, we assess the performance of five sets of classifier models using different feature selection techniques (SHAP-XGBoost vs. XGBoost, SHAP-DT vs. DT, SHAP-CatBoost vs. CatBoost, SHAP-ET vs. ET, and SHAP-RF vs. RF) with their respective selected features. The top 3, 5, 7, 10, and 15 features are selected based on their respective scores. For classification, we build credit card fraud detection models using the five classifiers, the same models used in feature selection. The classifiers are evaluated using the Area Under the Precision Recall Curve (AUPRC) metric [13], and we additionally perform a statistical test with a significance level of $\alpha = 0.01$ to assess the statistical significance of our results.

To the best of our knowledge, this study is the first comprehensive empirical investigation comparing the performance of SHAP-value-based feature selection and importance-based feature selection in the context of fraud detection and potentially other application domains in machine learning.

The remainder of the paper is organized as follows. We begin with an overview of related work, which shows the novelty of the research work we exhibit here. Following that we present the methodology used in the experiment, including explanations of two feature methods, classifiers, cross-validation, and performance metric. We then describe the datasets, experimental design, and experimental results. Finally, we conclude the article with key highlights of this study, and offers suggestions for future work.

Related work

Feature selection is a widely used technique in various data mining and machine learning applications. Its primary objective is to identify a subset of features that minimizes prediction errors for classifiers. In this study, we conducted a comprehensive literature review of research that employs either SHapley Additive exPlanations (SHAP) values or the model's built-in feature importance list for feature selection. While we found a limited number of studies that utilized the model's built-in feature importance list for feature selection in the context of the Credit Card Fraud Detection Dataset, we did not come across any studies that used SHAP for feature selection specifically in credit card fraud detection. Instead, we found a few studies that applied SHAP for feature selection in other application domains. Moreover, we did not encounter any studies that directly compared the performance of models built with features selected by SHAP feature importance versus models built with features selected by built-in feature importance. Therefore, our study presents a unique contribution to the field of credit card fraud detection, as it explores the comparison between SHAP and the model's built-in feature importance list for feature selection, a perspective that has not been extensively explored in the existing literature.

Rtayli and Enneya [14] applied a supervised feature selection method, Random Forest, to identify the most predictive features. Random Forest (RF) is an ensemble learning algorithm that is trained in parallel through bagging [15]. Recently, RF has been increasingly exploited as a feature selection method because it can handle complex, high-dimensional datasets and can detect interactions between features. It also reduces the risk of overfitting, which occurs when a model is too complex and fits the training data too closely. Moreover, RF calculates the feature importance by measuring the decrease in the impurity of the node when the feature is used for the split. The more the impurity decreases, the more important the feature is considered. By ranking the features based on their importance, RF can help select the most relevant features for the classification task. After selecting a feature subset from the Credit Card Fraud Detection Dataset, the authors ran Support Vector Machine to find fraudulent transactions. The model achieved an Accuracy of 95.12%, a Sensitivity of 87%, and an AUC of 0.91, outperforming three other models (Isolation Forest, Decision Tree, and Local Outlier Factor). The study does not provide clear information regarding the number of selected features. Additionally, the authors did not conduct a comparison of the performance between the selected features and the usage of all the available features. Furthermore, it is worth noting that the use of AUC as a metric for classification of imbalanced data has been found to be misleading [16].

In their study using the Credit Card Fraud Detection Dataset [12], Rosley et al. [17] first filtered out the data with a z-score greater than or equal to 3 and then normalized the remaining data using min-max scaling. Then they used Boruta to compute the importance score of each feature. Boruta [18] is a supervised feature selection algorithm that is designed as a wrapper around a Random Forest classifier to identify important features in a dataset. They kept the features with an importance score of 0.5 or higher to train the Autoencoder for each iteration. The model detected credit card fraud by defining a threshold in the reconstruction error to flag the transactions as legitimate or fraudulent. However, the number of features selected in the

preprocessing step has not been specified by the authors. The authors evaluated the models using Accuracy, Precision, Recall, and F1 score. When working with datasets that exhibit significant class imbalance, these may not be suitable metrics due to the overwhelming size of the majority class.

Waspada et al. [4] use the RF classifier to calculate the importance score of each feature. Features with a low importance score are discarded. The paper lists the importance score of all features. The authors analyze several factors (dataset split ratio, the selection of top k features, the amount of fraud data on training data, and the setting of hyper-parameter values) that influence the performance of the Isolation Forest (IF) model to detect fraud on credit card transactions. Isolation Forest is a popular unsupervised outlier detection method. Their findings indicate that the best results can be obtained by setting training–testing ratio of 60:40, using the top five features (V_{14} , V_4 , V_{17} , V_{12} , V_{11}), using only 60% of fraud data, and setting hyper-parameters with the number of trees 100, 128 sample maximum, and 0.001 contamination. The model shows impressive results obtaining precision of 80.7143%, recall of 76.3514%, F1 score of 78.4722%, Area Under the Receiver Operating Characteristic Curve (AUC) of 0.97371, and Area under the Precision-Recall Curve (AUPRC) of 0.759228. Waspada et al. utilized only a single importance-based feature selection method and did not incorporate SHAP for feature selection, which we have implemented in our study.

In their study, Liu et al. [19] utilized SHAP for feature selection on the UCI Parkinson's disease medical dataset [20]. They combined SHAP values with four classifiers: Deep Forest (gcForest), Extreme Gradient Boosting (XGBoost), Light Gradient Boosting machine (LightGBM), and Random Forest (RF). Each classifier was used to calculate the SHAP values of individual features. To assess the effectiveness of SHAP feature selection, they compared it with three filter-based feature selection methods: Fscore, analysis of variance (Anova-F), and Mutual Information. The experiments were conducted with a training and testing ratio of 70:30, and the feature selection was applied to the training dataset. The results showed that the gcForest model based on SHAP value feature selection achieved an impressive classification Accuracy of 91.78% and an F1-score of 0.945, with 150 features selected. This performance surpassed the outcomes of other feature selection methods considered in their study. While the authors specifically employed SHAP-value-based feature selection on the training dataset, we utilized the SHAP method across the entire dataset and subsequently conducted cross-validation following the feature selection procedure.

Marcilio and Eler [21] employed the SHAP method as a feature selection technique and compared it against three widely used feature selection methods: Mutual Information, Recursive Feature Elimination, and ANOVA. The SHAP process involved utilizing XGBoost as the underlying model. They conducted experiments on five UCI datasets using the XGBoost classifier and three other UCI datasets using the XGBoost regressor. The results of their study revealed that SHAP outperformed the three commonly used methods in terms of the Area Under the Receiver Operating Characteristic Curve (AUC) metric. However, it was observed that SHAP required more computational time compared to the other feature selection methods. It is worth noting that the datasets used in Marcilio and Eler's experiments are not highly

imbalanced, and not in the credit card fraud domain. In addition, the datasets are significantly smaller in size compared to the Kaggle Credit Card Fraud Detection Dataset, which caught our attention.

In our review of the literature, we discovered that only a single method of feature selection, either based on SHAP values or importance, was employed. Notably, no research has been identified that compares these two methods, particularly within the domain of credit card fraud detection. In order to fill this gap, our study undertook a comparative analysis of these two feature selection methods, employing five learners in each approach.

Methodology

Importance-based feature selection methods

Importance-based feature selection methods leverage decision trees to identify relevant features from a given dataset. These decision tree-based classifiers, such as Extreme Gradient Boosting (XGBoost) [6, 22], Extremely Randomized Trees (ET) [9], Random Forest (RF) [23], CatBoost [8], and Decision Tree [7], possess a built-in capability to determine feature importance during model fitting in supervised machine learning. Consequently, they can rank features based on their significance in classification tasks, making them valuable for feature selection. By discarding less relevant features and retaining the most important ones, more efficient and accurate models can be created.

In this study, five importance-based feature selection methods were employed: XGBoost [22], Decision Tree (DT) [7], CatBoost [8], Extremely Randomized Trees (ET) [9], and Random Forest (RF) [10].

XGBoost and CatBoost stand out as widely used gradient boosting algorithms, each employing distinct approaches to compute feature importance scores. While both algorithms construct ensembles of decision trees, their methodologies for deriving feature importance scores vary. In XGBoost, these scores are calculated using the “gain” method, evaluating the influence of each feature on model performance throughout the boosting process. In contrast, CatBoost’s ensemble of decision trees calculates feature importance based on the frequency of a feature being utilized for splitting and the subsequent improvement in model performance achieved through those splits.

A Decision Tree classifier is a type of machine learning algorithm used for classification tasks. It constructs a tree-like model of decisions and their potential outcomes by recursively splitting the data based on the most informative features at each node. Decision trees generate feature importance scores by evaluating their ability to reduce Gini impurity (or increase purity) within the data as the tree is built.

Extremely Randomized Trees and Random Forest, both rooted in decision tree ensembles, share common principles like Gini impurity and the Mean Decrease in Impurity to gauge feature importance. However, Extremely Randomized Trees introduce heightened randomness in the decision-making process during tree construction. This added stochasticity can result in divergent importance scores, potentially impacting the balance between model bias and variance.

SHAP-value-based feature selection methods

Shapley Additive exPlanation (SHAP), introduced by Lundberg and Lee [3], has gained popularity as a method for interpreting machine learning model predictions. By utilizing Game Theory techniques [24], SHAP provides insights into the contribution of each feature to specific predictions. It falls under a family of additive feature attribution techniques that remain model-agnostic, making them universally applicable to various machine learning and deep learning models. These techniques attribute significance to individual input features, facilitating better understanding of model behavior.

In the context of feature selection, SHAP-based methods work as follows: classification models, such as XGBoost and Decision Tree in this study, are trained on the entire dataset. Subsequently, SHAP values are computed for each instance, and these values are then aggregated across the dataset to derive average absolute values for each feature. The computation of SHAP values becomes computationally complex due to this process. The average SHAP value indicates the typical impact of each feature on model predictions across the entire dataset, while the absolute SHAP value represents the feature's importance, irrespective of its direction (positive or negative). By sorting features based on their average absolute SHAP values in descending order, features with higher SHAP values are identified as more influential in influencing the model's predictions.

Classification

In this study, credit card fraud detection models were built with five different classifiers, namely XGBoost [6], Decision Tree (DT) [7], CatBoost [8], Extremely Randomized Trees (ET) [9], and Random Forest (RF) [10]. Among these five learners, XGBoost, CatBoost, ET, and RF are ensemble of Decision Tree-based classifiers [25]. We select these learners on the basis that they are highly effective for dealing with complex, high-dimensional data and are known for their excellent performance in a wide range of classification tasks [25].

XGBoost and CatBoost are all gradient boosting frameworks that are widely used for machine learning tasks, particularly for classification. These two algorithms are known to be highly effective and produce accurate predictions. However, the performance may vary depending on the specific dataset and problem at hand. XGBoost is an advanced refinement the Gradient Boosted Decision Tree (GBDT) ensemble method. GBDTs were initially introduced by Friedman in 2001 [26]. XGBoost enhances GBDTs in multiple ways. Firstly, it employs an improved loss function during training that includes an additional term for regularization, effectively preventing overfitting. Secondly, XGBoost introduces an "approximate algorithm" for calculating splits in the constituent decision trees, which is highly suitable for distributed environments and cases where the entire dataset cannot fit into main memory. Moreover, XGBoost incorporates a specialized algorithm for handling sparse data, where most values are nearly constant with occasional aberrations. The "sparsity aware split finding" feature enables XGBoost to capitalize on sparse data efficiently. CatBoost, on the other hand, is known for its robustness in handling categorical features and missing values, making it suitable for datasets with such characteristics. CatBoost's core algorithm is Ordered Boosting, which involves sorting the instances used by Decision

Trees. In contrast, XGBoost relies on a weighted quantile sketch and a function that takes into account sparsity. A weighted quantile sketch is an approximate tree learning [27] technique that is utilized for merging and pruning operations, while sparsity deals with values that are either zero or missing.

Breiman introduced the concept of Bagging in the domain of machine learning in a 1996 paper [28]. As our research revolves around binary classification, our focus is on Breiman's ideas about Bagging applied to binary classification. Extremely Randomized Trees (ET) and Random Forest (RF) are both ensemble learning algorithms that belong to the bagging family of decision tree-based methods. Random Forest, which was introduced by Breiman [10]. Random Forest builds upon the Bagging principle with an added improvement. In a Random Forest, each tree is constructed using a random subset of features and samples. This randomness helps to decorrelate the trees and reduce overfitting. Extremely Randomized Trees extends the concept of Random Forest by selecting values for Decision Tree splits at random, potentially making them more robust and computationally efficient in some scenarios. The choice between the two often depends on the specific characteristics of the data and the desired trade-off between bias and variance. We skip the detailed information about these learners and readers are referred to [25].

Decision Tree (DT) is a widely used supervised machine learning algorithm, prominently applied to classification and regression tasks. It is a non-linear model that recursively partitions input data into subsets based on feature values. Each node in the decision tree represents a decision based on a specific feature and threshold, facilitating predictions based on the input data's feature values. The resulting decision tree structure is highly interpretable, with each internal node representing a feature-based decision, edges signifying outcomes, and leaf nodes providing predictions.

To ensure the reproducibility of our results, we modified specific hyperparameter settings from their default values as listed in Table 1. Furthermore, we set random number generator seeds for all classifiers to ensure consistent and repeatable outcomes. All other settings were left at their default values. The determination of tree depths was guided by previous experimentation documented in [1], aiming to achieve a suitable trade-off between capturing complex patterns in the data and mitigating overfitting.

Table 1 Hyperparameter settings used in experiments

Classifier	Parameter name	Parameter setting
CatBoost	task_type	GPU*
	max_ctr_complexity	1
	max_depth	5
ET	max_depth	8
XGBoost	max_depth	3
	tree_method	gpu_hist*
Random Forest	max_depth	4

* Setting selects Graphics Processing Unit (GPU) implementation of the classifier

Performance metric

To assess the effectiveness of feature selection techniques, we constructed classification models subsequent to the feature selection process. The evaluation of these models in this study was based on the Area under the Precision-Recall Curve (AUPRC) metric.

In a two-class classification problem, such as distinguishing fraud (positive) and normal (negative) instances, we encounter four potential prediction outcomes: true positive (correctly classified positive instances), false positive (negative instance mistakenly classified as positive), true negative (correctly classified negative instances), and false negative (positive instance mistakenly classified as negative).

AUPRC represents the area under the Precision-Recall curve, which illustrates the trade-off between Recall (True Positive Rate) and Precision for specific classification thresholds. The definition of precision is

$$\frac{\text{true positives}}{\text{true positives} + \text{false positives}} \quad (1)$$

and the Recall or True Positive Rate is defined as

$$\frac{\text{true positives}}{\text{true positives} + \text{false negatives}} \quad (2)$$

To calculate AUPRC, we plot precision against recall for many classification thresholds and then determine the area under the curve. A higher AUPRC value indicates superior model performance. AUPRC ranges from a minimum of zero to a maximum of one.

Cross-validation

Cross-validation refers to a technique used to allow for the training and testing of machine learning models without resorting to using the same data [29]. The process involves dividing the dataset into a predetermined number of subsets or folds in a relatively balanced manner. In this study, we utilized five-fold cross-validation, where each fold served as the test data, while the remaining four folds were designated as the training data. To minimize any potential bias arising from a fortuitous or unfavorable split, we conducted ten independent runs of the five-fold cross-validation.

It is important to note, for reproducibility, that the feature selection process was conducted separately from the cross-validation step. In other words, the feature selection procedures were performed on the original dataset.

Experiments

Dataset

The experiments conducted in this study utilized the Credit Card Fraud Detection Dataset, which is available for download from the Kaggle website [12]. This dataset consists of anonymized financial transactions, specifically credit card transactions conducted by European cardholders over a two-day period in September 2013. As stated previously, out of a total of 284,807 transactions, 492 of them are

fraudulent transactions, resulting in an imbalanced dataset with only 0.172% of transactions being fraudulent, while the rest are considered normal or non-fraudulent transactions.

The Credit Card Fraud Detection Dataset has 30 numerical input features, out of which $V_1, V_2, \dots, V_{28}$ have undergone numerical transformation using Principal Component Analysis (PCA) for data analysis and feature reduction purposes. However, the “Time” and “Amount” features were not transformed. The “Time” feature denotes the time in seconds since the first transaction, while the “Amount” feature represents the amount of the credit card transaction. The “Time” feature was excluded from the analysis to avoid influencing the reliability of the results since it is a unique feature that a model can memorize. As a result, there are 29 input features available for further experimentation. Prior to being input to the classifiers for training or classification, the features were normalized to fit within the $[0, 1]$ range. The class feature is utilized to distinguish between legitimate and fraudulent transactions. In this context, a value of 1 represents a fraudulent transaction, while a value of 0 signifies a normal transaction.

Experimental design

In our experiments, we investigated two different feature selection techniques, SHAP-value-based feature selection and importance-based feature selection methods. To assess the efficacy of a feature selection method, we constructed classification models utilizing the subset of features chosen by the feature selection approach. Classification models were built with five classifiers, XGBoost, Decision Tree (DT), CatBoost, Extremely Randomized Trees (ET), and Random Forest (RF).

We conducted our experiments on a distributed computing platform consisting of nodes equipped with 16-core Intel Xeon CPUs, 256 GB RAM per CPU, and Nvidia V100 GPUs. All training and testing programs were implemented using the Python programming language. SHAP is publicly available as an open source library for the Python programming language [30]. In addition to the SHAP values for feature importance, this library also supplies several tools for visualizing SHAP feature importance values. The Python data science stack [31] was employed for experiment implementations.

First, we ranked the features using ten feature selection methods (SHAP-XGBoost, XGBoost, SHAP-DT, DT, SHAP-CatBoost, CatBoost, SHAP-ET, ET, SHAP-RF, and RF) separately. Following feature ranking, we chose the top 3, 5, 7, 10, and 15 features, including the class attribute, to construct the final training datasets. Subsequently, we applied classifiers to these training datasets, ensuring that the classifier used in the model-building process remained consistent with the one employed in feature selection. We used AUPRC to evaluate the performance of the classification models. For each feature selection method and classifier, we have a total of 5 (feature subset sizes) $\times 10$ (runs) $\times 5$ (folds) = 250 AUPRC scores.

Results and discussion

As mentioned earlier, we have introduced ten feature selection methods, two feature selection techniques combined with five classifiers. We present the feature importance lists obtained from each method, where we focus on the top 15 most important features. The importance is determined either by SHAP values (for SHAP-XGBoost, SHAP-DT,

Table 2 Features selected by SHAP-XGBoost and XGBoost; the features are listed in order of their importance values from top to bottom

Ranking	SHAP-XGBoost	XGBoost
1	V_{14}	V_{17}
2	V_4	V_{14}
3	V_{12}	V_{10}
4	Amount	V_{27}
5	V_8	V_{12}
6	V_{11}	V_{26}
7	V_7	V_4
8	V_{10}	V_1
9	V_5	V_8
10	V_{19}	V_7
11	V_{26}	V_{16}
12	V_{27}	V_9
13	V_3	Amount
14	V_{16}	V_{13}
15	V_{18}	V_3

Table 3 Features selected by SHAP-DT and DT; the features are listed in order of their importance values from top to bottom

Ranking	SHAP-DT	DT
1	V_{14}	V_{17}
2	V_{17}	V_{14}
3	V_{12}	V_{27}
4	V_4	V_{12}
5	V_1	V_{10}
6	V_{20}	V_{26}
7	V_{19}	V_{24}
8	V_8	V_{16}
9	V_{10}	V_7
10	V_7	V_{20}
11	V_{21}	V_4
12	V_{26}	V_1
13	V_{27}	V_{23}
14	Amount	V_{19}
15	V_{22}	V_{15}

SHAP-CatBoost, SHAP-ET, and SHAP-RF) or built-in importance scores (for XGBoost, DT, CatBoost, ET, and RF). In Tables 2, 3, 4, 5, 6, we display the feature rankings, where rank 1 corresponds to the highest SHAP value or importance score. It's important to note that SHAP values may vary when different trained models are utilized. Notably, among all ten feature selection methods, feature V_{14} stood out as one of the top three features. Additionally, feature V_4 consistently appeared and held a ranking within the top 15 across all feature selection methods.

The classification performance results in terms of AUPRC are shown in Tables 7, 8, 9, 10, 11. The reported values represent averages across ten rounds of five-fold cross-validation outcomes. The results were obtained by creating new datasets using the 3,

Table 4 Features selected by SHAP-CatBoost and CatBoost; the features are listed in order of their importance values from top to bottom

Ranking	SHAP-CatBoost	CatBoost
1	V_1	V_1
2	V_{14}	V_4
3	V_4	V_{14}
4	V_8	Amount
5	V_{26}	V_{11}
6	V_6	V_{26}
7	Amount	V_{13}
8	V_{24}	V_8
9	V_{12}	V_{17}
10	V_{13}	V_3
11	V_{11}	V_{20}
12	V_{18}	V_{18}
13	V_{10}	V_{24}
14	V_{17}	V_{15}
15	V_{19}	V_{28}

Table 5 Features selected by SHAP-ET and ET; the features are listed in order of their importance values from top to bottom

Ranking	SHAP-ET	ET
1	V_{14}	V_{14}
2	V_{17}	V_{17}
3	V_{12}	V_{12}
4	V_4	V_{10}
5	V_{10}	V_{11}
6	V_{11}	V_{16}
7	V_{16}	V_{18}
8	V_3	V_4
9	V_9	V_9
10	V_{18}	V_3
11	V_7	V_7
12	V_{19}	V_2
13	V_1	V_{21}
14	V_2	V_{19}
15	V_{15}	V_{26}

5, 7, 10, and 15 highest-ranked features along with the class attribute to form the final training data. We conducted statistical z -tests [32] on pairs of models (same classifier but different feature selection methods), where each pair consists of one model built with n of the most important features selected by SHAP or the model's built-in feature importance list. The value of n ranges from 3 to 15. The null hypothesis is that there is no significant difference between the mean AUPRC scores of the two models. In Tables 7, 8, the Winner column indicates whether the SHAP or built-in feature selection method has a higher mean AUPRC value based on the outcome of a z -test with a significance level of $\alpha = 0.01$. If the difference in means is not significant, we report a tie.

Table 6 Features selected by SHAP-RF and RF; the features are listed in order of their importance values from top to bottom

Ranking	SHAP-RF	RF
1	V ₁₄	V ₁₇
2	V ₁₇	V ₁₂
3	V ₁₂	V ₁₄
4	V ₁₀	V ₁₀
5	V ₄	V ₁₆
6	V ₁	V ₁₁
7	V ₁₁	V ₉
8	V ₁₆	V ₄
9	V ₂	V ₇
10	V ₇	V ₁₈
11	V ₁₉	V ₂₆
12	V ₃	V ₂₁
13	V ₅	V ₁
14	Amount	V ₈
15	V ₁₈	V ₃

Table 7 Comparison of SHAP and XGBoost feature selection methods in terms of their AUPRC scores

Size	SHAP-XGBoost	XGBoost	p-value	Winner
3	0.7247	0.7727	0.0000	XGBoost
5	0.8165	0.7978	0.0121	Tie
7	0.8302	0.8255	0.5005	Tie
10	0.8446	0.8350	0.0041	SHAP-XGBoost
15	0.8535	0.8557	0.7097	Tie

Table 8 Comparison of SHAP and DT feature selection methods in terms of their AUPRC scores

Size	SHAP-DT	DT	p-value	Winner
3	0.7421	0.7323	0.4968	Tie
5	0.7493	0.7414	0.6293	Tie
7	0.7594	0.7666	0.7013	Tie
10	0.7380	0.7686	0.2429	Tie
15	0.7664	0.7564	0.5058	Tie

Table 9 Comparison of SHAP and CatBoost feature selection methods in terms of their AUPRC scores

Size	SHAP-CatBoost	CatBoost	p-value	Winner
3	0.6106	0.7235	0.0000	CatBoost
5	0.7266	0.7745	0.0000	CatBoost
7	0.7897	0.8279	0.0000	CatBoost
10	0.8333	0.8472	0.0000	CatBoost
15	0.8506	0.8491	0.7502	Tie

Table 10 Comparison of SHAP and ET feature selection methods in terms of their AUPRC scores

Size	SHAP-ET	ET	<i>p</i> -value	Winner
3	0.7796	0.7843	0.6756	Tie
5	0.8172	0.8118	0.4243	Tie
7	0.8143	0.8137	0.9179	Tie
10	0.8175	0.8168	0.9152	Tie
15	0.8086	0.8048	0.7238	Tie

Table 11 Comparison of SHAP and RF feature selection methods in terms of their AUPRC scores

Size	SHAP-RF	RF	<i>p</i> -value	Winner
3	0.8097	0.8137	0.5673	Tie
5	0.8396	0.8248	0.0133	Tie
7	0.8416	0.8382	0.6126	Tie
10	0.8447	0.8479	0.6399	Tie
15	0.8544	0.8512	0.6693	Tie

Table 12 ANOVA for Size, Classifier and Technique as factors of performance in terms of AUPRC

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Size	4	1.90	0.48	237.44	less than 10^{-4}
Classifier	4	2.11	0.53	262.91	less than 10^{-4}
Technique	1	0.05	0.05	24.77	less than 10^{-4}
Residuals	2490	4.99	0.00		

Table 7 shows a tie for XGBoost models built on feature subset sizes of 5, 7, and 15. However, for feature subset size 3, the *p*-value is less than the significance level of 0.01, indicating a significant difference in the AUPRC scores. Therefore, XGBoost outperforms SHAP-XGBoost for feature count 3. On the other hand, for feature subset size 10, SHAP-XGBoost outperforms XGBoost.

Table 8 indicates that there is no significant difference in the AUPRC scores between SHAP-DT and DT for any of the feature counts tested (3, 5, 7, 10, and 15). As a result, we cannot declare a winner between the two feature selection methods based on the AUPRC scores. Tables 10 and 11 are similar to Table 8. The results suggest that, for the given dataset and evaluation metric, there is no consistent superior performance between the SHAP feature selection methods and the traditional importance-value based decision tree, extra tree, or random forest methods across different feature sizes.

Table 9 presents a comparison between SHAP-CatBoost and CatBoost feature selection methods in terms of their AUPRC scores for different feature sizes. In summary, for feature sizes 3–10, CatBoost consistently outperforms SHAP-CatBoost in terms of AUPRC, and the differences are statistically significant with *p*-values of 0.0000. However, for size 15, there is no statistically significant difference between the two methods, resulting in a tie.

In general, the performance of the two feature selection methods is comparable across various scenarios. However, there are specific instances, such as with certain XGBoost and CatBoost models, where distinctions arise. Notably, XGBoost demonstrates superior performance over SHAP-XGBoost when the feature subset size is 3, while CatBoost outperforms SHAP-CatBoost for feature sizes 3, 5, 7, and 10. Moreover, SHAP-XGBoost surpasses XGBoost when the feature subset size is 10.

An analysis of variance (ANOVA) [33] was performed on AUPRC performance metrics, and the results are reported in Table 12. Three factors, Size, Classifier, and Technique, were considered in the analysis. The Size Factor included feature subset sizes of 3, 5, 7, 10, and 15, the Classifier Factor included five classifiers, while the Technique factor included two feature selection methods, SHAP-value based (Represented with SHAP) and Importance-value based (represented with Importance). The statistical test used a significance level of $\alpha = 1\%$. The ANOVA results indicate that there were significant differences among the groups in each of the main factors in terms of the AUPRC metric, as all $\Pr(>F)$ or p-values in the last column of the table were less than the cutoff of 0.01.

Since the ANOVA test results revealed that all factors had a significant impact on AUPRC scores, we conducted Tukey’s Honestly Significant Difference (HSD) tests [34] to rank the Technique and Classifier based on their impact on AUPRC scores. The performance was ranked alphabetically, with group ‘a’ having the highest AUPRC scores. Items in the same performance group indicate no statistically significant difference between them. The HSD test results are presented in Tables 13, 14, 15.

Based on the HSD tests, it is evident that feature selection with a subset size of 15 and 10 yields superior performance in AUPRC compared to smaller subset sizes. This suggests that constructing models with a feature subset size of 15 or 10 is advantageous. The reduced size leads to faster model training times and improved outcomes. Among the five classifiers, RF demonstrated the highest AUPRC, followed by XGBoost and ET,

Table 13 HSD test groupings after ANOVA of AUPRC for the Size factor

Group a consists of: 15
Group ab consists of: 10
Group b consists of: 7
Group c consists of: 5
Group d consists of: 3

Table 14 HSD test groupings after ANOVA of AUPRC for the Classifier factor

Group a consists of: RF
Group b consists of: XGBoost, ET
Group c consists of: CatBoost
Group d consists of: DT

Table 15 HSD test groupings after ANOVA of AUPRC for the Technique factor

Group a consists of: Importance
Group b consists of: SHAP

while DT showed relatively poorer performance. Table 15 indicates that the importance-value-based feature selection method significantly outperforms the SHAP-value-based feature selection method, across all feature subsets sizes, and learners.

As mentioned earlier, SHAP is an external tool, and the computational time for SHAP feature selection depends on several factors, including the model's complexity, the number of features, the dataset size, and the number of instances for which SHAP values need to be computed. The complexity of computing SHAP values is generally higher than other feature importance methods like decision-tree-based classifiers. Therefore, we conclude that using the built-in feature importance to select feature subsets may be more suitable for models with a large number of features and a large dataset.

Conclusion

The challenge of dealing with high dimensionality in machine learning significantly affects the evaluation of model performance. This study specifically concentrates on the comparison of two feature selection techniques: identifying the most crucial features through SHAP values and relying on the model's intrinsic feature importance list. Using the Credit Card Fraud Detection Dataset, we generate multiple training datasets. We employ five classifiers with distinct feature subset sizes, applying both feature selection methods to each classifier. Our results indicate that, on the whole, feature selection methods based on importance values outperform those based on SHAP values across the classifiers used in this study and various feature subset sizes.

However, notable variations arise in XGBoost models. XGBoost surpasses SHAP-XGBoost for a feature subset size of 3, while SHAP-XGBoost outperforms XGBoost for a feature subset size of 10. In the case of CatBoost, CatBoost outperforms SHAP-CatBoost for feature sizes less than 15. It is important to note that calculating SHAP feature importance introduces an additional step in the experimental methodology. According to our findings, the return on investment for implementing SHAP may be relatively low, particularly when built-in feature selection methods are available, especially for large datasets. Additionally, the considerable computational expenses associated with SHAP may render it impractical for handling Big Data. For future research, our plan is to explore these two feature selection methods across diverse application domains.

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Author contributions

HJW contributed to the manuscript. QXL and JTH conducted experiments, and JTH contributed to the manuscript. TMK provided oversight of experiments, coordinated research, and contributed to the manuscript.

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